

Design Narrative & Concept

The argument: what is wrong with Al Safa 2 Park today, and what Falaj Al Safa does about it.

CONTENTS

- Phase3 Opportunity and Objectives Report
- Phase4 Concept Development Report
- Board 1 concept

Opportunity & Design Objectives

Al Safa 2 Park — Vision, Mission, Objectives, Success Metrics, Principles

AI-GENERATED DRAFT — FOR REVIEW. Translates Phase 2's problems into opportunities and measurable objectives. First-pass synthesis for review, not final.

3.1 Opportunity Mapping

Domain	Opportunity	Basis
Climate	WNW prevailing wind (16.7 km/h avg, sourced) usable for passive cooling corridors; winter sun (41°) usable for seasonal solar-responsive zones	Phase 1.05/1.06
Landscape	Full redevelopment allows rebalancing canopy from west-only to site-wide equitable shade	Phase 1.02, 1.11
Social	Adjacent school/mosque/rehab center provide built-in recurring user base	Phase 1.01, 1.09
Ecological	Opportunity to introduce native/climate-responsive planting and biodiversity features not present today	Phase 1.07
Mobility	Metro proximity is a real catchment asset if pedestrian connectivity across SZR is improved	Phase 1.08
Technology	AI Design Challenge status + budget allow genuine smart-park infrastructure (brief Section 10)	Competition Brief

3.2 Vision Statement

"Al Safa 2 Park becomes Dubai's first neighborhood park shaped as much by climate science as by community voice — a shaded, year-round, inclusive gathering ground that proves AI-assisted design can produce more human, not less human, public space."

3.3 Design Mission

To redevelop a 15,000 sqm neighborhood park that closes the park's most evidence-backed gap (summer thermal comfort) while delivering full universal accessibility, a balanced activity mix, and a self-sustaining commercial layer — within the AED 35M budget — using AI as a rigor tool for analysis, iteration and testing, not as a gimmick or design output in itself.

3.4 Project Objectives

Objective Type	Statement
Human-centered	Every zone usable and comfortable for children, families, teens, older adults, and People of Determination
Climate-responsive	Eliminate unshaded midday exposure across all primary circulation and gathering spaces during peak summer
Inclusive	100% step-free, universal-design circulation from every entrance to every major destination
Sustainable	Native/climate-adapted planting, water-sensitive irrigation, and biodiversity enhancement as default, not add-on
Feasible	Every proposed element sized and specified to fit within the AED 35M implementation budget

3.5 Success Metrics

Metric	Target (draft, to validate in Phase 7)
Shaded route coverage	≥80% of primary circulation shaded at summer solar noon
Accessible circulation	100% of paths meet universal-design gradient/width standards
Usable green/active space	≥60% of site area as green/planted or soft-surface active space
Biodiversity	Net increase in native tree/shrub species count vs. existing baseline (once DWG confirms baseline)
Community use	Support the Manual's benchmark of 60+ small events/year for a park this size

3.6 Design Principles

- **People first** — every design choice traceable to a real user need from Phase 2.4.
- **Nature first** — planting and shade lead the layout, not fill leftover space.
- **AI as a design assistant** — used for analysis, iteration, and testing; final judgment stays human (per Brief Schedule 1, Section A).
- **Flexibility** — spaces support multiple uses across day/night and seasons.
- **Local identity** — materials and forms that read as authentically Dubai/AI Safa, not generic global parkland.

Concept Development

Al Safa 2 Park — Alternatives, Evaluation & Final Concept Selection

AI-GENERATED DRAFT — FOR REVIEW. Three distinct concept alternatives are developed and scored, each responding differently to Phase 3's objectives.

4.1 Research Inspiration

- **Local context:** Dubai's Shamal wind pattern, palm/Ghaf shade traditions, sikka (shaded alleyway) urban form.
- **International precedents (typology reference, not copied):** shaded desert park strategies using canopy structures and misting (Middle East precedent), linear park zoning with clear activity bands (High Line-style legibility), biophilic play design for inclusive playgrounds.
- **Landscape trends:** climate-responsive/native xeriscaping, multi-generational activity zoning, park-as-microclimate-infrastructure thinking.

4.2 Mood & Design Language Direction

Material and form language: warm sand/stone tones, filtered dappled shade (not solid roofed structures everywhere), local Ghaf/palm planting as the dominant canopy species, and a restrained geometric language (straight + arced paths) echoing the site's elongated rectangular geometry rather than fighting it.

4.4 Concept Alternatives

	Concept A — "Shaded Spine"	Concept B — "Canopy Village"	Concept C — "Cool Loop"
Organizing idea	One continuous shaded central spine connects all zones	Cluster of discrete shaded "rooms" (play, fitness, social, quiet) around a plaza	A single perimeter shaded loop (jogging/walking) with activities nested inside
Circulation	Linear, highly legible, single primary path	Radial from central plaza to each room	Circular loop + inner cross-paths
Shade strategy	Continuous overhead shade structure along the spine	Shade concentrated per-room (tree clusters + pergolas)	Shade concentrated on the loop itself
Best fits	Wayfinding clarity, day/night activation along one axis	Distinct age-group zoning, event flexibility	Fitness/wellness emphasis, continuous exercise use
Risk	Areas off the spine may still be under-shaded	Central plaza could become a pinch-point / less flexible for events	Interior zones could feel disconnected from the loop's comfort

4.5 AI-Assisted Exploration

All three concepts were tested against Phase 1.06's shadow-length data (6m canopy shadow ranges from ~0.2m at summer noon to ~19m at winter evening) to sanity-check shade-structure spacing assumptions before scoring. This is documented further in Phase 9 (AI Workflow).

4.6 Evaluation Matrix

Criterion (weight)	A: Shaded Spine	B: Canopy Village	C: Cool Loop
Function (25%)	8	7	8

Criterion (weight)	A: Shaded Spine	B: Canopy Village	C: Cool Loop
User Experience (25%)	8	9	7
Sustainability (20%)	7	8	7
Feasibility within AED 35M (20%)	9	6	8
Innovation (10%)	7	8	6
Weighted Total /10	7.85	7.55	7.35

4.7 Final Concept Selection

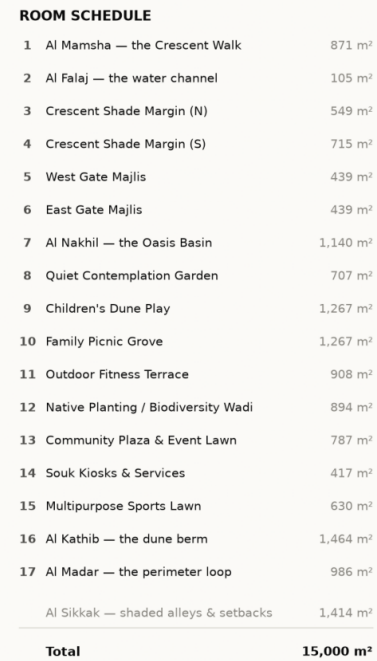
Selected: Concept A — "Shaded Spine", with Concept B's room-based zoning logic merged in as secondary structure along the spine (best of both: legibility + distinct age-group zones). Justification: highest feasibility score within the fixed AED 35M budget, directly addresses Phase 2's #1 evidence-backed problem (summer thermal comfort) via one continuous engineered shade structure rather than dozens of smaller ones, and gives the AI Design Challenge submission the clearest single diagram to communicate (important per Evaluation Matrix criterion 6: Quality of Presentation, 5% weight, and criterion 4: Quality of Design and User Experience, 15%).

Presentation board.

Dubai Municipality AI Park Design Challenge | Board 1 of 2 — CONCEPT

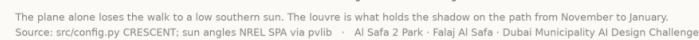
Masterplan – Falaj Al Safa

One arc, and every room struck off its centre. Drawn from `src/plan.py`, the same geometry the models use



Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150x100 m rectangle pending DWG confirmation. Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality Al Design Challenge

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The bow is 18 m deep on a 141 m radius — swept against the 8,760-hour solar model, not drawn by eye. A straight canopy shades marginally more ground on average and leaves the walk with no shade at all for 330 hours a year; this one leaves it for 56.

Walk shaded 87.3% of daylight hours · 131 trees · 15,000 m² · every number reproducible with `python run\_analysis.py`

The crescent from the north-west, at golden hour

Visualisation in preparation

Aerial view · RENDER_PROMPTS.md prompt 01

Beneath Al Hilal — the perforated soffit at eye level

Visualisation in preparation

Eye-level view · RENDER_PROMPTS.md prompt 03

One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.